

Problem 1:

S.N	line	Error Description	Error Type	Error Correction
1	8	Missing semicolon after <code>int x = 10</code>	Syntax Error	Add ; at the end of the statement
2	1 2	Assignment operator <code>=</code> used in condition <code>if(x = 10)</code>	Logical Error	Use comparison operator <code>==</code>
3	1 7	String "Ninety" assigned to integer variable <code>marks</code>	Type Mismatch Error	Assign an integer value
4	1 9	Variable <code>z</code> used without declaration	Semantic Error	Declare and initialize <code>z</code> before use
5	2 1	Cannot assign value to constant (<code>100 = x</code>)	Syntax Error	Assign value to a variable instead
6	2 5	Array index 10 exceeds size of <code>arr[5]</code>	Array Out of Bounds	Use index within range 0-4
7	2 7	Address of integer variable assigned to <code>float *ptr</code>	Type Mismatch Error	Use compatible pointer type
8	2 9	<code>%f</code> format specifier used with integer variable <code>x</code>	Format Specifier Error	Use <code>%d</code> for integer values
9	3 4	Division by zero (<code>a/b</code> , where <code>b = 0</code>)	Runtime Error	Ensure divisor is not zero
10	3 6	Multi-character constant 'AB' assigned to <code>char</code>	Semantic Error	Use a single character constant
11	3 9	Both <code>if</code> and <code>else</code> print the same message	Logical Error	Print appropriate message in <code>else</code> block
12	4 4	Function <code>show()</code> called without declaration	Compilation Error	Declare function prototype before use
13	4 4	Function <code>show()</code> not defined	Linker Error	Provide function definition
14	4 6	Semicolon after <code>while(x < 15)</code> creates an empty loop	Logical Error	Remove unnecessary semicolon

15	5 1	<code>continue</code> used outside any loop	Syntax Error	Use <code>continue</code> only inside loops
16	5 3	<code>break</code> used outside any loop/switch	Syntax Error	Use <code>break</code> only inside loop or switch
17	5 5	Function <code>calculate()</code> called with one argument instead of two	Function Argument Error	Pass both required arguments
18	3 4	Program may terminate due to divide-by-zero	Runtime Error	Handle divisor validation before division

Problem 2:

Line	Error Description	Correction
8	Missing semicolon after <code>int num2 = 20</code>	Add <code>;</code> at the end of the statement.
19	Missing closing parenthesis in <code>if(choice == 1</code>	Add <code>)</code> "before the opening brace <code>{</code> .
30	Extra closing parenthesis in <code>else if(choice == 3))</code>	Remove the extra <code>)</code> .
45	Missing closing parenthesis in <code>for(int i=0; i<5; i++</code>	Add <code>)</code> before the opening brace <code>{</code> .
52	Missing closing parenthesis in <code>while(choice > 0</code>	Add <code>)</code> before the opening brace <code>{</code> .
60	Missing semicolon after <code>while(choice > 5)</code> in do-while	Add <code>;</code> after the condition.
67	Missing colon <code>(:)</code> after <code>case 2</code>	Add <code>:</code> after the case label.
92	Missing closing quote in <code>char grade = 'A;</code>	Close the character constant properly with <code>'</code> .
96	Missing closing bracket in <code>int arr[5;</code>	Add <code>]</code> before the semicolon.
121	Missing semicolon after <code>printf(...)</code>	Add <code>;</code> at the end.
102-119	Opening block <code>{</code> is not properly closed	Add required closing brace <code>}</code> .

Line	Error Description	Error Type	Correction
3	Function <code>add()</code> declared but never used	Warning	Call the function or remove its declaration.
38	Integer division performed in <code>num1 / num2</code>	Logical Warning	Use a <code>float</code> or <code>double</code> if decimals are needed.
20-23	Addition performed directly instead of using <code>add()</code>	Logical Warning	Use the <code>add()</code> function if that was the intent.
55	While loop only decrements; no visible output	Logical Warning	Add meaningful processing inside the loop.
57-60	Do-while loop condition may be incorrect	Logical Error	Verify the loop condition logic.

Line	Error Description	Correction
92	Unterminated character literal	Use a valid, closed character constant.
96	Invalid array declaration	Complete the declaration syntax.
102-119	Nested block structure is unbalanced	Ensure all opened blocks are properly closed.

Problem 3:

S.No.	Line	Error Description	Error Type	Error Correction
1	8	String "twenty" assigned to integer variable <code>age</code>	Type Mismatch	Assign an integer value
2	14	String "A" assigned to char variable <code>grade</code>	Type Mismatch	Use a single character constant
3	18	<code>%d</code> format specifier used with float variable <code>salary</code>	Format Specifier	Use <code>%f</code>

4	18	%f format specifier used with integer variable <code>marks</code>	Format Specifier	Use %d
5	22	Variable <code>score</code> used without declaration	Semantic Error	Declare and initialize <code>score</code>
6	25	Variable <code>x</code> redeclared in the same scope	Compilation Error	Remove duplicate or rename
7	21	Cannot assign value to constant (<code>100 = x</code>)	Syntax Error	Assign value to a variable instead
8	31	%s format specifier used with integer variable <code>num</code>	Format Specifier	Use %d
9	35	%d format specifier used with float variable <code>pi</code>	Format Specifier	Use %f
10	38	Array index <code>2.5</code> is not an integer	Array Index Error	Use an integer index

S.No.	Line	Error Description	Error Type	Error Correction
11	41	Float value assigned to char variable <code>ch</code>	Type Conversion	Use valid character data
12	45	Function <code>add()</code> called with one argument instead of two	Function Argument	Pass both required arguments
13	51	Function <code>average()</code> called with two arguments instead of three	Function Argument	Pass all required arguments
14	55	String "25" assigned to integer variable <code>temperature</code>	Type Mismatch	Assign an integer value
15	58	String "10.5" assigned to float variable <code>discount</code>	Type Mismatch	Assign a numeric float value
16	67	Direct assignment <code>name = "Rishi"</code> to character array	Compilation Error	Use string copy function
17	73	Variable <code>totalMarks</code> used without declaration	Semantic Error	Declare and initialize <code>totalMarks</code>
18	81	%f format specifier used with float variable <code>rate</code>	Format Specifier	Use %f (Correcting existing)
19	85	Variable <code>c</code> used without declaration	Semantic Error	Declare <code>c</code> before use
20	89	Function <code>add()</code> called with three arguments instead of two	Function Argument	Pass only two arguments

21	103	Assignment operator used in (studentMarks = 100)	Logical Error	Use comparison operator ==
22	109	String "Large" assigned to integer variable size	Type Mismatch	Assign an integer value
23	115	%d format specifier used with float variable percentage	Format Specifier	Use %f
24	118	String "five" used as array index	Array Index Error	Use an integer index

S.No.	Line	Error Description	Error Type	Error Correction
25	128	String constant "2" used in case label of integer switch	Type Mismatch	Use integer constant 2
26	136	Function multiply() called without declaration/definition	Compilation/Linker	Declare and define the function
27	148	%d format specifier used with float variable areaRect	Format Specifier	Use %f
28	154	Return value of average() (float) assigned to integer avgMarks	Type Conversion	Use float variable or explicit cast
29	160	Variable value used without declaration	Semantic Error	Declare and initialize value
30	112	String "EMP101" assigned to integer variable employeeId	Type Mismatch	Use a string variable
31	164	%d format specifier used for string based employeeId	Format Specifier	Use %s with string variable

Problem 4:

S.No.	Line No.	Error	Error Type	Error Correction
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1	26	Function <code>calculateSalary()</code> declared but not defined	Linker Error	Provide function definition
2	28	Function <code>printStudentDetails()</code> declared but not defined	Linker Error	Provide function definition
3	34	Function <code>showResult()</code> declared but not defined	Linker Error	Provide function definition
4	36	Function <code>calculateAverage()</code> declared but not defined	Linker Error	Provide function definition
5	44	Function <code>generateReport()</code> called without declaration	Compilation Error	Declare function prototype before use
6	44	Function <code>generateReport()</code> not defined	Linker Error	Provide function definition
7	46	Function <code>displayEmployee()</code> called without declaration	Compilation Error	Declare function prototype before use
8	46	Function <code>displayEmployee()</code> not defined	Linker Error	Provide function definition
9	48	Function <code>processData()</code> called without declaration	Compilation Error	Declare function prototype before use
10	48	Function <code>processData()</code> not defined	Linker Error	Provide function definition
11	50	Function <code>saveRecord()</code> called without declaration	Compilation Error	Declare function prototype before use
12	50	Function <code>saveRecord()</code> not defined	Linker Error	Provide function definition
13	52	Function <code>loadRecord()</code> called without declaration	Compilation Error	Declare function prototype before use
14	52	Function <code>loadRecord()</code> not defined	Linker Error	Provide function definition
15	54	Function <code>printInvoice()</code> called without declaration	Compilation Error	Declare function prototype before use
16	54	Function <code>printInvoice()</code> not defined	Linker Error	Provide function definition
17	56	Function <code>calculateTax()</code> called without declaration	Compilation Error	Declare function prototype before use
18	56	Function <code>calculateTax()</code> not defined	Linker Error	Provide function definition
19	58	Function <code>validateUser()</code> called without declaration	Compilation Error	Declare function prototype before use
20	58	Function <code>validateUser()</code> not defined	Linker Error	Provide function definition
21	60	Function <code>sendNotification()</code> called without declaration	Compilation Error	Declare function prototype before use
22	60	Function <code>sendNotification()</code> not defined	Linker Error	Provide function definition
23	62	Function <code>generateCertificate()</code> called without declaration	Compilation Error	Declare function prototype before use

24	62	Function <code>generateCertificate()</code> not defined	Linker Error	Provide function definition
25	40	<code>companyProfit</code> declared as extern but no definition provided	Linker Error	Define <code>companyProfit</code> globally
26	15	Variable <code>choice</code> declared but never used	Warning	Remove variable or use it in program body

Problem 5:

S.No.	Line No.	Error	Error Type	Error Correction
1	11	Division by zero (a / b , where $b = 0$)	Runtime Error	Ensure divisor is not zero before division
2	15	NULL pointer <code>ptr1</code> dereferenced	Runtime Error	Assign valid memory before dereferencing
3	19	Array index 10 exceeds size of <code>arr[5]</code>	Array Out of Bounds Error	Use valid indices 0-4
4	23	Uninitialized pointer <code>ptr2</code> dereferenced	Runtime Error	Initialize pointer before use
5	27	String "Programming" exceeds size of <code>name[5]</code>	Buffer Overflow Error	Increase array size or shorten string
6	33	Uninitialized pointer <code>ptr3</code> dereferenced	Runtime Error	Initialize pointer before dereferencing
7	39	<code>fscanf()</code> used without checking if file opened successfully	Runtime Error	Verify <code>fp != NULL</code> before reading
8	45	String "Computer Science" exceeds allocated memory of <code>ptr4</code>	Buffer Overflow Error	Allocate sufficient memory
9	51	Accessing memory after <code>free(ptr4)</code>	Use After Free Error	Do not access freed memory
10	57	Double free of <code>ptr5</code>	Runtime Error	Free memory only once
11	63	Modulo by zero ($x \% y$, where $y = 0$)	Runtime Error	Ensure divisor is not zero
12	69	<code>strcat()</code> causes buffer overflow in <code>str1[10]</code>	Buffer Overflow Error	Ensure destination array has enough space

13	77	Writing 10 elements into memory allocated for only 3 integers	Buffer Overflow Error	Allocate sufficient memory or reduce loop limit
14	85	Accessing matrix[5][5] outside 3x3 array bounds	Array Out of Bounds Error	Use valid row and column indices
15	89	NULL pointer ptr6 dereferenced	Runtime Error	Check pointer before access
16	93	Use of gets() function	Unsafe Function Error	Use a safer input function
17	103	Writing to memory after free(ptr7)	Use After Free Error	Do not access freed memory
18	107	Variable age used without initialization	Uninitialized Variable Error	Initialize variable before use
19	111	Negative array index arr[-1]	Array Out of Bounds Error	Use valid non-negative indices
20	115	Uninitialized pointer ptr8 dereferenced in condition	Runtime Error	Initialize pointer before use
21	123	Printing NULL string pointer text	Runtime Error	Check pointer before printing
22	129	malloc() return value not checked before use	Runtime Error	Verify allocation success before access
23	141	Infinite loop while(1)	Logical Error	Add termination condition
24	39	File "unknown.txt" may not exist	Runtime Error	Check file existence before reading
25	75	Dynamically allocated memory numbers may be corrupted due to overflow	Runtime Error	Stay within allocated bounds
26	127	Extremely large memory allocation may fail	Memory Allocation Error	Validate allocation result
27	51	Dangling pointer used after memory release	Runtime Error	Set pointer to NULL after free()
28	103	Dangling pointer ptr7 used after free()	Runtime Error	Avoid accessing freed

Problem 6:

S.No.	Line No.	Error	Error Type	Error Correction
1	15	Condition <code>num % 2 == 1</code> prints "Number is Even"	Logical Error	Print "Odd" for odd numbers
2	17	Else block prints "Number is Odd"	Logical Error	Print "Even" for even numbers

3	25	Marks $\geq$ 90 assigned Grade B	Logical Error	Assign correct highest grade
4	27	Marks $\geq$ 80 assigned Grade A	Logical Error	Correct grade sequence
5	48	Age > 18 prints "Not Eligible for Voting"	Logical Error	Print "Eligible for Voting"
6	50	Else block prints "Eligible for Voting"	Logical Error	Print "Not Eligible for Voting" when required
7	59	Larger number printed as smaller number	Logical Error	Print actual smaller value
8	61	Else block prints larger value as smaller	Logical Error	Print actual smaller value
9	72	Comparison uses < while finding largest number	Logical Error	Compare correctly for largest value
10	75	Comparison uses < while finding largest number	Logical Error	Compare correctly for largest value
11	87	Integer division in average calculation	Logical Error	Use floating-point division
12	97	Formula used is circumference-related, not area	Logical Error	Use correct circle area formula
13	109	Bonus percentage 0.02 instead of intended higher rate	Logical Error	Apply correct bonus rate
14	121	Month 1 mapped to February	Logical Error	Map to correct month
15	125	Month 2 mapped to March	Logical Error	Map to correct month
16	129	Month 3 mapped to April	Logical Error	Map to correct month
17	133	Month 4 mapped to May	Logical Error	Map to correct month
18	148	<code>isPrime</code> remains 1 when factor found	Logical Error	Set flag appropriately
19	154	Prime number printed as "Not Prime"	Logical Error	Print correct result
20	156	Non-prime number printed as "Prime"	Logical Error	Print correct result
21	168	Discount stored as fixed value 5 instead of percentage/amount logic	Logical Error	Apply correct discount calculation
22	172	Final amount calculated using addition	Logical Error	Subtract discount from purchase amount
23	183	<code>sum = i</code> overwrites previous sum	Logical Error	Accumulate values properly

24	195	Temperature < 40 prints "Cold Weather"	Logical Error	Use appropriate temperature condition
25	205	Grade A prints "Average Performance"	Logical Error	Print appropriate performance message
26	217	Rectangle area calculated using addition	Logical Error	Use multiplication
27	227	Percentage $\geq$ 60 prints "Fail"	Logical Error	Print "Pass"
28	229	Else block prints "Pass"	Logical Error	Print "Fail" when required

Problem 7:

S.No.	Line No.	Error	Error Type	Error Correction
1	8	Loop condition <code>i <= 5</code> accesses <code>marks[5]</code>	Array Out of Bounds Error	Use <code>i < 5</code>
2	24	Integer division used in <code>average = total / 5</code>	Logical Error	Use type casting for accurate floating-point result
3	32	Loop starts from index 1, leaving <code>numbers[0]</code> uninitialized	Logical Error	Start loop from index 0
4	32	Loop condition <code>i <= 10</code> accesses <code>numbers[10]</code>	Array Out of Bounds Error	Use valid indices 0 to 9
5	46	String "ComputerScience" exceeds size of <code>name[10]</code>	Buffer Overflow	Increase array size or shorten string
6	54	Direct assignment <code>city = "Delhi"</code> to character array	Compilation Error	Use string copy function
7	63	String comparison using <code>==</code>	Logical Error	Use string comparison function
8	78	Use of <code>gets()</code> function	Unsafe Function Error	Use a safer input function with size limit
9	84	<code>strcat()</code> may overflow <code>firstName[20]</code>	Buffer Overflow	Ensure destination array has sufficient space
10	94	Character array <code>grade</code> not null terminated before printing as string	Runtime / Logical Error	Add string terminator <code>'\0'</code>

11	102	Negative array index <code>arr[-1]</code>	Array Out of Bounds Error	Use valid index range
12	110	<code>%s</code> with <code>scanf()</code> may overflow <code>department[10]</code>	Buffer Overflow Risk	Limit input size

S.No.	Line No.	Error	Error Type	Error Correction
13	118	String "student@gmail.com" exceeds size of <code>email[15]</code>	Buffer Overflow Error	Increase array size
14	127	Writing to <code>scores[3]</code> when array size is 3	Array Out of Bounds Error	Use valid indices 0 to 2
15	129	Reading <code>scores[3]</code> outside array bounds	Array Out of Bounds Error	Access only valid indices
16	133	String "Programming" exceeds size of <code>subject[10]</code>	Buffer Overflow Error	Increase array size or shorten string
17	143	String comparison <code>username == "admin"</code>	Logical Error	Use string comparison function
18	155	String "987654321012" exceeds size of <code>mobile[10]</code>	Buffer Overflow Error	Increase array size
19	168	Loop condition <code>i <= 5</code> accesses <code>attendance[5]</code>	Array Out of Bounds Error	Use <code>i < 5</code>
20	178	Accessing <code>course[20]</code> outside valid index range	Array Out of Bounds Error	Use index within array limits
21	190	Character array <code>college</code> not null terminated	Runtime / Logical Error	Add <code>'\0'</code> at the end
22	210	String "IT-A" exceeds size of <code>section[3]</code>	Buffer Overflow Error	Increase array size to store full string and terminator
23	30-40	Printing <code>numbers</code> array may display garbage value at index 0	Logical Error	Initialize all array elements properly
24	56	Assignment to array name is not allowed in C	Compilation Error	Copy string contents instead of assigning
25	118	Buffer overflow causes memory corruption in email array	Runtime Error	Allocate sufficient storage
26	156	Mobile number string requires space for null terminator	Buffer Overflow Error	Allocate enough characters including <code>'\0'</code>

Section 3: Error 27

S.No.	Line No.	Error	Error Type	Error Correction
27	84	Concatenated string length may exceed destination capacity	Runtime Error	Check combined length before concatenation

Problem 8:

S.No.	Line No.	Error	Error Type	Error Correction
1	15	Function <code>add()</code> called with one argument instead of two	Function Argument Error	Pass both required arguments
2	21	Function <code>average()</code> called with two arguments instead of three	Function Argument Error	Pass all three required arguments
3	61	Function <code>calculateGrade()</code> called without declaration or definition	Compilation / Linker Error	Declare and define the function
4	67	Function <code>circleArea()</code> called without declaration or definition	Compilation / Linker Error	Declare and define the function
5	75	Function <code>sumArray()</code> called without declaration or definition	Compilation / Linker Error	Declare and define the function
6	81	Function <code>displayResult()</code> called without declaration or definition	Compilation / Linker Error	Declare and define the function
7	97	Base case of factorial returns 0 for <code>n == 0</code>	Logical Error	Return 1 for factorial of 0
8	117	<code>maximum()</code> returns smaller value when <code>a > b</code>	Logical Error	Return the larger value
9	119	<code>maximum()</code> returns smaller value in else block	Logical Error	Return the larger value
10	132	Recursive call <code>fibonacci(n)</code> causes infinite recursion	Logical Error	Use smaller arguments in recursive calls

11	145	Integer division in percentage calculation may produce incorrect result	Logical Error	Use proper arithmetic/type casting
12	155	Non-void function <code>simpleInterest()</code> uses <code>return</code> without value	Compilation Error	Return the calculated value
13	170	Loop condition <code>i < exp</code> gives incorrect result for exponentiation	Logical Error	Ensure multiplication occurs required number of times
14	187	Function <code>reverseNumber()</code> has no return statement	Compilation Warning/Error	Return the reversed number
15	209	<code>isPrime()</code> returns 1 when number is divisible	Logical Error	Return 0 for non-prime numbers
16	212	<code>isPrime()</code> returns 0 when number is prime	Logical Error	Return 1 for prime numbers
17	226	Loop condition <code>i <= size</code> accesses one element beyond array size	Array Out of Bounds Error	Use <code>i < size</code>
18	233	Recursive function has no terminating condition	Runtime Error (Stack Overflow)	Add a valid base condition
19	243	Rectangle area calculated using addition	Logical Error	Use multiplication for area calculation
20	253	<code>cube()</code> returns <code>n * n</code> instead of cube value	Logical Error	Return cube of the number
21	37	<code>swap(x, y)</code> does not swap original variables	Logical Error	Use pointers or return swapped values
22	40	Output after swap remains unchanged	Logical Error	Modify original variables through references/pointers
23	91	Average calculation returns integer division result	Logical Error	Use floating-point division
24	132	Fibonacci implementation recalculates current term recursively	Logical Error	Use correct recursive formula
25	145	Percentage formula may truncate decimal values	Logical Error	Perform floating-point calculation
26	210	<code>updateValue()</code> modifies local copy only	Logical Error	Use pointer parameter to modify caller variable
27	233	Infinite recursion can exhaust program stack	Runtime Error	Add termination condition
28	226	Array printing function may access invalid memory location	Runtime Error	Restrict access to valid array indices

Problem 9:

S.No.	Line	Error Description	Error Type	Error Correction
1	13	Uninitialized pointer <code>ptr1</code> dereferenced	Runtime Error	Initialize pointer before dereferencing
2	17	NULL pointer <code>ptr2</code> dereferenced	Runtime Error	Assign valid memory address before use
3	25	Pointer increment moves <code>ptr3</code> beyond intended variable	Logical Error	Avoid accessing memory not belonging to the object
4	27	Dereferencing incremented pointer may access invalid memory	Runtime Error	Access only valid memory locations
5	33	Accessing <code>*(ptr4 + 10)</code> beyond array bounds	Array Out of Bounds Error	Use valid array indices
6	43	Accessing <code>*ptr5</code> after <code>free(ptr5)</code>	Runtime Error (Use After Free)	Do not access memory after freeing it
7	51	Allocated memory address lost when <code>ptr6 = NULL</code>	Memory Leak	Free memory before assigning NULL
8	67	<code>malloc</code> allocates only 5 bytes, insufficient for 10 integers	Memory Allocation Error	Allocate enough memory using <code>sizeof(int)</code>
9	81	Writing 10 elements into insufficient allocated memory	Buffer Overflow Error	Allocate adequate memory or reduce writes
10	73	Accessing <code>ptr8[3]</code> when only 2 elements allocated	Array Out of Bounds Error	Use valid indices 0 to 2
11	85	Double free of <code>ptr9</code>	Runtime Error	Free allocated memory only once
12	91	Reading uninitialized dynamically allocated memory <code>*ptr10</code>	Undefined Behavior	Initialize memory before reading
13	99	Loop condition <code>i <= 5</code> writes to <code>numbers[5]</code>	Array Out of Bounds Error	Use <code>i < 5</code>
14	115	<code>pptr = &value;</code> assigns <code>int*</code> to <code>int**</code>	Type Mismatch Error	Assign address of an integer pointer
15	117	Dereferencing invalid double pointer <code>**pptr</code>	Runtime Error	Ensure proper pointer levels and initialization
16	122	Allocated size 5 bytes is too small for "Programming"	Buffer Overflow Error	Allocate sufficient memory for string and null terminator

17	126	<code>strcpy()</code> used without including <code><string.h></code>	Compilation Warning/Error	Include required header file
18	135	Incrementing pointer <code>ptr11++</code> loses original allocated address	Memory Leak	Preserve original pointer
19	137	Dereferencing incremented pointer outside allocated object	Runtime Error	Access only allocated memory
20	141	<code>NULL</code> pointer <code>ptr12</code> dereferenced in condition	Runtime Error	Check pointer before dereferencing
21	153	Accessing <code>ptr13[100]</code> beyond allocated range	Array Out of Bounds Error	Use indices 0 to 99
22	161	Writing to memory after <code>free(ptr14)</code>	Runtime Error (Use After Free)	Do not access freed memory
23	167	Assigning 2D array to <code>int*</code> causes incompatible pointer type	Type Mismatch Warning	Use appropriate pointer type
24	169	Accessing <code>*(p + 20)</code> outside matrix bounds	Array Out of Bounds Error	Access only valid matrix elements
25	175	First allocated memory lost when <code>ptr15</code> is reassigned	Memory Leak	Free previous allocation before reassignment
26	185	Swap function swaps local pointer copies instead of values	Logical Error	Swap values pointed to by pointers
27	193	Loop condition <code>i <= size</code>	Array Out of Bounds	Use <code>i < size</code>

Problem 10=

S.No.	Line No.	Error	Error Type	Error Correction
1	10	Missing semicolon after <code>int choice = 1</code>	Syntax Error	Add <code>;</code> at the end of the statement
2	14	String "90" assigned to integer variable <code>marks</code>	Type Mismatch Error	Assign an integer value instead of a string
3	20	Function <code>add()</code> called with one argument instead of two	Function Argument Error	Pass both required arguments
4	24	Assignment operator (<code>=</code>) used in <code>if(x = 5)</code>	Logical Error	Use comparison operator <code>==</code>

5	28	Variable <code>count</code> used without declaration	Semantic Error	Declare and initialize <code>count</code> before use
6	31	Array index 10 exceeds size of <code>array_a[5]</code>	Array Out of Bounds Error	Use valid index range 0–4
7	33	Negative array index <code>arr[-1]</code>	Array Out of Bounds Error	Use non-negative valid index

S.No.	Line No.	Error	Error Type	Error Correction
8	38	String "Programming Language" exceeds size of <code>name[10]</code>	Buffer Overflow Error	Increase array size or shorten string
9	45	<code>char city = "Delhi"</code> assigned to character array	Compilation Error	Use <code>strcpy</code> copy function
10	52	String comparison using <code>==</code>	Logical Error	Compare string contents using <code>strcmp</code> function
11	60	Division by zero (<code>x/y</code> where <code>y = 0</code>)	Runtime Error	Ensure divisor is not zero
12	64	NULL pointer <code>ptr1</code> dereferenced	Runtime Error	Assign valid memory before dereferencing
13	68	Uninitialized pointer <code>ptr2</code> dereferenced	Runtime Error	Initialize pointer before use
14	75	Accessing <code>*ptr3</code> after <code>free(ptr3)</code>	Runtime Error (Use After Free)	Do not access memory after freeing it
15	80	Double free of <code>ptr3</code>	Runtime Error	Free allocated memory only once
16	84	<code>malloc(5)</code> allocates insufficient memory for 10 integers	Memory Allocation Error	Allocate memory using required size
17	88	Writing 12 elements into insufficient allocated memory	Buffer Overflow Error	Allocate enough memory or reduce loop range
18	94	Memory allocated to <code>ptr5</code> lost due to reassignment	Memory Leak	Free previous allocation before reassignment
19	100	Function <code>average()</code> called with two arguments instead of three	Function Argument Error	Pass all required arguments
20	105	Age greater than 18 prints "Not Eligible"	Logical Error	Print appropriate eligibility message

S.No.	Line No.	Error	Error Type	Error Correction
21	120	<code>isPrime</code> remains 1 even when divisor is found	Logical Error	Set <code>isPrime = 0</code> when number is divisible
22	124	Prime number printed as "Not Prime"	Logical Error	Print correct result based on <code>isPrime</code> value
23	130	Function <code>displayReport()</code> declared but not defined	Linker Error	Provide function definition
24	133	Function <code>calculateSalary()</code> called without declaration	Compilation Error	Declare function prototype before use
25	137	Function <code>calculateSalary()</code> not defined	Linker Error	Provide function definition
26	142	<code>fopen()</code> used without checking whether file opened successfully	Runtime Error	Verify file pointer is not NULL
27	148	Missing colon (:) after <code>case 1</code>	Syntax Error	Add : after case label
28	157	Multi-character constant 'AB' assigned to <code>char</code>	Semantic Error	Use a single character constant
29	167	Writing to <code>scores[3]</code> when array size is 3	Array Out of Bounds Error	Use valid indices 0–2
30	171	Loop condition <code>i <= 3</code> accesses <code>scores[3]</code>	Array Out of Bounds Error	Use <code>i < 3</code>
31	181	Formula for circle area is incorrect (<code>2 * pi * r</code>)	Logical Error	Use correct area formula
32	185	Semicolon after <code>while(count < 5)</code> creates empty loop	Logical Error	Remove unnecessary semicolon
33	195	<code>continue</code> used outside loop	Syntax Error	Use <code>continue</code> only inside loops
34	201	Average function performs integer division	Logical Error	Use floating point division for accurate average

35	145	Runtime Error Check the opening status before reading
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36 150 Logical Error Add `break` if fall-through is not intended